Energy Transport in Stellar Interiors

- radiation
- convection

<u>Radiation</u>: diffusive process

Diffusion is described by Fick's Law:
when there is a gradient in the density of particles, the diffusive flux: $\vec{J}$ is

$$\vec{J} = -D\nabla n \quad \text{where } D = \frac{1}{3} v_{ave}\, l$$

→ the flux of particles per area per time depends on the average particle velocity and their mean free path

Similarly, if there is a gradient in the energy density carried by the particles, then there is a net flux of energy crossing the surface:

$$\vec{F} = -D\nabla u$$

The gradient for u is due to the temperature gradient, so

$$\nabla u = \left(\frac{\partial u}{\partial T}\right)_V \nabla T = C_V \nabla T$$

$$\Rightarrow F = -D\nabla u = -\frac{1}{3} v_{ave}\, l\, C_V \nabla T = -K\nabla T$$

$$K = \frac{1}{3} v_{ave}\, l\, C_V \equiv \text{conductivity}$$

The radiative diffusion energy accounts for photons. In this case

$$v_{ave} = c \quad \text{and} \quad u = aT^4$$

Then, the specific heat is just

$$C_v = \frac{du}{dT} = 4aT^3$$

The photon mean free path is

$$l_{photon} = \frac{1}{\kappa\rho} \quad \text{where } \kappa : \text{opacity}$$

Then, we can calculate the radiative conductivity as

$$K_{rad} = \frac{1}{3} v_{ave} \, l \, C_v$$
$$= \frac{4}{3} \frac{caT^3}{\kappa\rho}$$

The radiative energy flux is then:

$$\vec{F}_{rad} = -K_{rad} \nabla T = -\frac{4}{3} \frac{aT^3 c}{\kappa\rho} \nabla T$$

If we consider the local luminosity, l, at radius r in a spherically symmetric star, the energy flux is one dimensional

$$F_{rad} = \frac{l}{4\pi r^2} \quad \text{and} \quad \nabla T \longrightarrow \frac{\partial T}{\partial r}$$

we can put everything together then

$$\frac{l}{4\pi r^2} = -\frac{4aT^3c}{3\rho\kappa}\frac{\partial T}{\partial r}$$, then solve for T gradient:

$$\frac{\partial T}{\partial r} = \frac{-3\kappa\rho}{16\pi aT^3c}\frac{l}{r^2}$$

$\longrightarrow$ this can be written in the mass coordinate using $\frac{\partial r}{\partial m} = \frac{1}{4\pi r^2\rho}$, then

$$\frac{\partial T}{\partial r} = 4\pi r^2\rho\frac{\partial T}{\partial m} = \frac{-3\kappa\rho}{16\pi aT^3c}\frac{l}{r^2}$$

$$\Rightarrow \boxed{\frac{\partial T}{\partial m} = \frac{-3}{64\pi^2 ac}\frac{\kappa l}{T^3r^4}}$$

If radiation is the dominant energy source, this is the temperature gradient required to carry the luminosity l to radius r by photons.

This equation is valid if the mean free path is much smaller than the size of the star.

We can use the assumption of hydrostatic equilibrium to write

$$\frac{dT}{dm} = \frac{dP}{dm}\frac{dT}{dP}, \text{ where } \frac{dP}{dm} = \frac{-Gm}{4\pi r^4}$$

Then, if we do derivative parkour again, we can write

$$\frac{d\log T}{d\log P} = \frac{P}{T}\frac{dT}{dP} \text{ and plug in.}$$

$$\frac{dT}{dm} = \frac{-Gm}{4\pi r^4}\frac{T}{P}\frac{d\log T}{d\log P} = \frac{-3}{64\pi^2 ac}\frac{\kappa l}{r^4 T^3}$$

$$\Longrightarrow \frac{d\log T}{d\log P} = \frac{3}{16\pi acG}\frac{\kappa l P}{mT^4} = \nabla_{rad}$$

This is the logarithmic temperature variation with P (which is a proxy for depth) for a star in hydrostatic equilibrium if energy is transported only through radiation

$\longrightarrow$ Skip mean opacity (5.2.3)

In the same way we consider photons colliding, we can consider gas collisions. However their cross sections are such that $l_{gas} \ll l_{photon}$.

$\Longrightarrow$ heat conduction is not as effective as radiative diffusion.

See Section 5.3 for a discussion of opacity

Sources: collisions between photons & electrons / ions

Eddington Luminosity

radiative transport of luminosity up through shells of a star requires a temperature gradient.

The radiative Pressure is

$$P_{rad} = \frac{1}{3} a T^4 \Rightarrow$$ a temperature grad implies a pressure grad

$$\rightarrow \frac{dP_{rad}}{dr} = \frac{4}{3} a T^3 \frac{dT}{dr} = \frac{4}{3} a T^3 \left[\frac{-3\kappa\rho}{16\pi a T^3 c} \frac{\ell}{r^2} \right]$$

$$= \frac{-\kappa\rho}{4\pi c} \frac{\ell}{r^2} = \frac{dP_{rad}}{dr}$$

$\longrightarrow$ photons apply an outward force

If the star is going to remain in hydrostatic equilibrium, the absolute pressure gradient from radiation must be Smaller than the pressure gradient enforced by HE.

$$\left| \frac{dP_{rad}}{dr} \right| < \left| \frac{dP_{HE}}{dr} \right| \Rightarrow \frac{\kappa\rho}{4\pi c} \frac{\ell}{r^2} < \frac{Gm\rho}{r^2}$$

$$\longrightarrow \ell < \frac{4\pi c}{\kappa} Gm \equiv \ell_{Edd}$$

at the surface: $L_{Edd} = \frac{4\pi c}{\kappa} GM$; κ: opacity in photosphere

For stars to remain in HE, $L < L_{edd}$. This means that either an extremely large Luminosity, due to huge heat flux, or a very high opacity can push a star out of HE.

High opacity happens at low temperatures
$\longrightarrow$ mode of transport in this case can't be radiative diffusion
(it's convection!)

High luminosity occurs when nuclear burning is strong. Assuming normal opacity, $\kappa = 0.34\ cm^2/g$,

$$L_{Edd} = \text{const} * \text{Mass} = 3.8 \times 10^4\ L_\odot \left(\frac{M}{M_\odot}\right)$$

$\longrightarrow$ there exists a maximum mass that doesn't exceed Ledd using M-L relation

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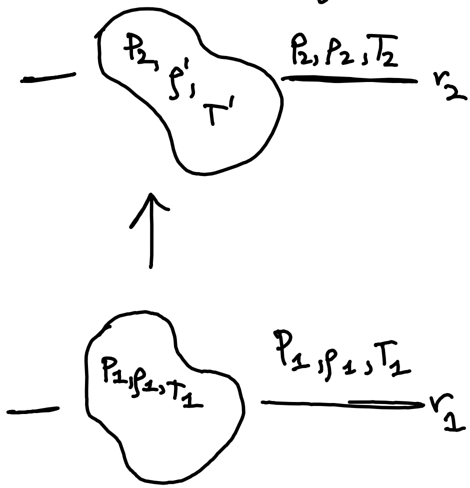
Convection: CONVECTION IS HARD!!!

We will write down a limiting case for convective instabilities or perturbations to grow.

Consider a gas blob that is moving adiabatically.

→ hydrostatic equilibrium is in play, but thermal equilibrium is not.



The direction the blob moves depends on the buyancy Forces.

If $T' > T_2$: blob is hotter than surrounding gas

The pressure is proportional to ρ and T, thus:

if $T' > T_2$: P_2 is fixed

$$\Longrightarrow \rho' < \rho_2$$

→ blob rises

For convection to develop, we require $T' > T_2$:

$$\frac{T' - T_1}{\Delta r} > \frac{T_2 - T_1}{\Delta r} \Longrightarrow \left(\frac{dT}{dr}\right)_{\text{blob}} > \left(\frac{dT}{dr}\right)_{\text{surrounding gas}}$$